

Register Number												
-----------------	--	--	--	--	--	--	--	--	--	--	--	--

VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Continuous Assessment Test - I		QP Set : I	Regulations : 2022
Programme : B.Tech - AI&DS	Semester : 3	Max. Marks : 60	Duration : 2 Hrs
Course Code & Title : 25ADT31 & FOUNDATION OF DATA SCIENCE AND ANALYTICS			
Class : II B.Tech - AI&DS		Date : 05.03.2026	Time : 10.30 am - 12.30 pm
Knowledge Levels (KL)	K1 – Remembering K2 – Understanding	K3 – Applying K4 – Analysing	K5 – Evaluating K6 – Creating

Part A – 12 × 2 = 24 Marks

Answer ALL questions.

No.	Question	CO	KL
1.	Define data science and its need.	CO1	K1
2.	List any five applications of data science.	CO1	K1
3.	State the purpose of setting a research goal in the data science process.	CO1	K1
4.	Illustrate a data transformation for a simple example.	CO1	K2
5.	Explain the concept of handling missing values with a suitable example.	CO1	K2
6.	Explain exploratory data analysis and its purpose.	CO1	K2
7.	Name the Python modules used in the data science process with their purpose.	CO2	K1
8.	Describe the purpose of data modeling in the data science process.	CO2	K2
9.	Explain the relation between probability and statistics.	CO2	K2
10.	Differentiate between descriptive and inferential statistics.	CO2	K2
11.	Distinguish between null hypothesis and alternative hypothesis with examples.	CO2	K2
12.	Calculate the standard error of the mean for the following dataset: 12, 14, 16, 18.	CO2	K3

Part B – 3 × 12 = 36 Marks

Answer ALL questions, choosing either (a) or (b) in each.

No.	Question	Marks	CO	KL
13. (a)	Explain the stages of the data science process with a neat diagram.	12	CO1	K2
(OR)				
(b)	Describe the steps involved in the data preparation phase and compare cleaning with transformation.	12	CO1	K4

No.	Question	Marks	CO	KL
14. (a)	Apply min-max normalization and z-score standardization to the data 10, 20, 30, 40, 50 and interpret the results.	12	CO2	K3
(OR)				
(b)	(i) Compute the mean and variance of the sample 10, 12, 14, 16.	6	CO2	K3
	(ii) Analyze the effect of an outlier on the mean and the median of the same sample.	6		K4
15. (a)	Evaluate the suitability of random sampling versus stratified sampling for a student survey and justify your answer.	12	CO3	K5
(OR)				
(b)	Design an end-to-end data science workflow to predict student performance from academic data.	12	CO3	K6

———— End of Question Paper ————

HoD Sign

Waiting for approval